

Computer Vision Systems CAP6411 Assignment#08

Ashmal Vayani, UCF ID: 5669011

Report: WAN2.1 on Human Activity Recognition Dataset

Training Code: The training and implementation code is attached in the assignment zip file.

1 Dataset: Human Activity Recognition (Videos)

We use the Human Activity Recognition (HAR) – Video Dataset released on Kaggle by Sharjeel Mazhar. The corpus contains short video clips grouped into seven activity classes: *Clapping*, *Meet and Split*, *Sitting*, *Standing Still*, *Walking*, *Walking While Reading Book*, and *Walking While Using Phone*. In total, the dataset includes approximately 1,113 clips spanning these categories. Videos are stored per class in separate directories and are typically around a few seconds long, commonly at 320×240 resolution (with durations and frame counts varying across clips). The dataset serves as a compact benchmark for human action recognition in videos.¹

Class	# Clips	Notes
Clapping	146	short hand clap sequences
Meet and Split	147	brief encounter, then separate
Sitting	156	subject sits / remains seated
Standing Still	174	static standing posture
Walking	171	normal pace walking
Walking While Reading Book	176	walking while reading
Walking While Using Phone	143	walking while looking at phone
Total	~1,113	seven classes overall

Table 1: HAR Video Dataset: classes and clip counts (counts with dashes should be filled from the local copy).

2 Model Architecture Details: Wan 2.1

Overview. Wan 2.1 is a family of diffusion–transformer video generative models available in **1.3B** and **14B** parameter scales. The models follow the diffusion–transformer (DiT) paradigm trained with a *flow-matching* objective, and pair a spatio–temporal VAE with a transformer denoiser conditioned on a multilingual text encoder. The suite also includes variants for T2V (text-to-video), I2V (image-to-video), FLF2V (first/last-frame to video) and VACE (controllable “any-to-video”).

Video VAE (Wan–VAE). Wan 2.1 introduces a *3D causal VAE* tailored to video. The encoder compresses spatio–temporal content into a latent volume while preserving temporal causality; the decoder reconstructs frames from these latents. The design emphasizes efficient spatio–temporal compression and low memory cost, and is reported to *encode/decode unlimited-length 1080p videos* without losing long-range temporal information. This VAE is a drop-in component for all Wan pipelines in **diffusers**.

¹See dataset page and prior usage: Kaggle (HAR–Video); counts reported in *Two-Stream Modality-Based Deep Learning Approach for HAR* and Kaggle notebooks.

Text conditioning. Prompts are embedded with a multilingual **UMT5** (T5) text encoder. Each DiT block applies *cross-attention* to inject text features into the visual tokens. Time (diffusion) embeddings are processed by a small MLP (Linear+SiLU) that predicts *six modulation parameters*; these parameters are shared via a single MLP across blocks while each block learns its own biases. This configuration is reported to improve performance at fixed parameter budgets.:contentReference[oaicite:2]index=2

Diffusion-Transformer (DiT) denoiser. The denoiser operates on the VAE latent video tokens using the flow-matching formulation within the standard DiT backbone. Architectural hyperparameters for the two public sizes are listed below (model/FFN dimensions, heads, and layers).

Table 2: Wan 2.1 DiT configuration (from the official model card).

Model	Dim	FFN Dim	# Heads	# Layers	Input Dim	Output Dim
T2V-1.3B	1536	8960	12	30	16	16
T2V-14B	5120	13824	40	40	16	16

Pipelines and modalities. The `diffusers` implementation exposes multiple pipelines: (i) **Wan-Pipeline** (T2V) using UMT5 + DiT + Wan-VAE; (ii) **WanImageToVideoPipeline** (I2V/FLF2V) which augments the stack with a CLIP vision encoder for frame conditioning; (iii) **WanVACE** for control signals (depth/pose/sketch/flow, inpainting/outpainting, subject/composition control, etc.).

Resource characteristics. The series targets both efficiency and scale. The 1.3B model is positioned for consumer GPUs (the project highlights ~ 8 GB VRAM use in their overview), while the 14B T2V diffusers example indicates ~ 13 GB VRAM to run the text-to-video pipeline with standard settings (with offloading options provided).

2.1 Training Objective and Sampling

Wan 2.1 adopts a **flow matching** objective within the DiT framework, replacing the classic DDPM noise-prediction loss. At inference, standard classifier-free guidance is used (exposed as `guidance_scale` in the pipelines). Flow-matching plus the 3D causal VAE are cited as the main enablers for fast, high-fidelity video synthesis.

Task-specific conditioning T2V: prompt text only (UMT5). **I2V/FLF2V:** prompt text + an image (and optionally last frame) encoded by CLIP-ViT; the DiT conditions on both text and image features and predicts the latent trajectory that reconstructs a temporally coherent clip via the VAE.

Wan 2.1 combines (i) a *3D causal* video VAE for efficient spatio-temporal latents, (ii) a *DiT* denoiser trained with *flow matching*, and (iii) *UMT5-based* multilingual text conditioning via per-block cross-attention and MLP-driven time-conditioning modulation. Public checkpoints cover **1.3B** and **14B** scales and multiple pipelines (T2V, I2V, FLF2V, VACE) in the Hugging Face `diffusers` ecosystem.

3 Report and Code

3.1 Errors and Obstacles Faced

- **Diffusers checkpoint layout mismatch.** Loading a locally downloaded Wan2.1-T2V-1.3B directory failed with `OSError: no file named pytorch_model.bin / model.safetensors`. The folder was not in the *Diffusers* layout (expected `text_encoder/`, `transformer/`, `vae/` plus a valid `model_index.json`). *Fix:* Use the official Diffusers-format release and point the pipeline to that path.²
- **Pipeline import confusion.** Importing a non-existent class (e.g., `WanT2VPipeline`) raised `ImportError`. *Fix:* Instantiate with `DiffusionPipeline.from_pretrained(<wan-dir>)`; Diffusers resolves the correct Wan pipeline internally.⁴

²<https://huggingface.co/Wan-AI/Wan2.1-T2V-1.3B-Diffusers>

³<https://huggingface.co/docs/diffusers/en/api/pipelines/wan>

⁴<https://huggingface.co/docs/diffusers/en/api/pipelines/wan>

- **LoRA integration.** Needed a proper Diffusers LoRA path. *Fix:* `pipe.load_lora_weights(path, adapter_name=...)` then `pipe.set_adapters([...], [...])`; optionally `pipe.fuse_lora()`.^{5 6}
- **Float-frame warnings when saving.** `imageio` printed: “Lossy conversion from float32 to uint8.” *Fix:* Convert frames with `(np.clip(f,0,1)*255).astype(np.uint8)` or use `diffusers.utils.export_to_video`.^{7 8}
- **Runtime speed / latency.** Default `num_inference_steps` and `num_frames` were slow. *Fix:* Reduce to `steps ≈ 28–36` and `frames ≈ 49–61`; enable TF32 math on Ampere: `torch.set_float32_matmul_precision("high")`.⁹
- **torch.compile + CUDA graphs crash.** Compiling the Wan transformer triggered: “*accessing tensor output of CUDAGraphs that has been overwritten.*” *Fix:* Disable Inductor CUDA graphs (or skip `torch.compile`); alternatively mark step boundaries when available.¹⁰
- **Generator device mismatch for seeding.** Drawing integer seeds from a CUDA `torch.Generator` threw: “*Expected a 'cpu' device type for generator.*” *Fix:* Sample seeds with a CPU generator, then create a device-specific generator for the pipeline call (`generator=...`).
- **Triton autotune OOM messages.** With `torch.compile`, Triton sometimes reported “*No valid triton configs ... OutOfMemoryError.*” *Fix:* Let it fall back to eager kernels or reduce compile scope / disable CUDA graphs.¹¹
- **Deprecated or benign warnings.** Replaced deprecated `torch_dtype` with `dtype`. The “*model type t2v to instantiate umt5*” warning is expected in Diffusers’ Wan composition.¹²
- **Filesystem robustness.** Some runs produced no outputs because the target folder didn’t exist. *Fix:* Guard all save paths with `os.makedirs(..., exist_ok=True)` and wrap `imageio.mimwrite` in `try/except` with explicit error prints.

3.2 Requirements and Dependencies

The exact `requirements.txt` used for our experiments is summarized below (CUDA build of PyTorch assumed).

```
requirements.txt

torch==2.4.1
torchvision==0.19.1
torchaudio==2.4.1
diffusers==0.35.0
transformers==4.45.2
accelerate==0.34.2
safetensors==0.4.4
huggingface-hub==0.24.6
tokenizers==0.19.1
sentencepiece==0.2.0
xformers==0.0.28.post3
numpy==1.26.4
scipy==1.13.1
pandas==2.2.2
imageio==2.34.1
imageio-ffmpeg==0.5.1
pillow==10.4.0
tqdm==4.66.5
```

⁵<https://huggingface.co/docs/diffusers/en/api/loaders/lora>

⁶https://huggingface.co/docs/diffusers/v0.25.1/using-diffusers/loading_adapters

⁷<https://stackoverflow.com/questions/58893860/lossy-conversion-from-float64-to-uint8>

⁸<https://huggingface.co/docs/diffusers/v0.25.0/en/api/utilities>

⁹https://docs.pytorch.org/en/stable/generated/torch.set_float32_matmul_precision.html

¹⁰<https://github.com/pytorch/pytorch/issues/144961>

¹¹<https://github.com/pytorch/pytorch/issues/144961>

¹²<https://huggingface.co/Wan-AI/Wan2.1-T2V-1.3B-Diffusers>

System prerequisites.

- NVIDIA GPU with CUDA support (tested on A100); recent driver compatible with CUDA 12.x.
- ffmpeg available on \$PATH (for MP4 writing via imageio-ffmpeg).
- Python 3.10 (tested).

Repro settings. Unless stated otherwise we used:

- dtype=bfloat16 on CUDA; TF32 enabled via `torch.set_float32_matmul_precision("high")`.
- Resolution **832×480**, **24 fps**, **num_frames=61**, **num_inference_steps=32**, **guidance_scale=6.0**.
- LoRA (optional): loaded with `pipe.load_lora_weights(...)` and scaled by `pipe.set_adapters(...)`.
- `torch.compile`: *disabled* by default to avoid CUDA graphs issues; can be enabled if desired.

3.3 CLI Commands to Reproduce

setup & run

```
(conda create env and install deps)
conda create -y -n cvs.ass8 python=3.10
conda activate cvs.ass8

(clone Wan2.1, this is needed for inference)
git clone https://github.com/Wan-Video/Wan2.1
cd Wan2.1
pip install -r requirements.txt

(clone diffusion-pipe for finetuning)
git clone https://github.com/tdrussell/diffusion-pipe/
cd diffusion-pipe
pip install -r requirements.txt

(training the model)
cd ./Assignments
sbatchrun.slurm

(inference)
(Run simple HAR generator (simple prompts))
python har_wan_generate_simple.py \
--ckpt_dir /path/to/Wan2.1-T2V-1.3B-Diffusers \
--size 832*480 \
--out_root ./videos/har_videos_simple \
--num_per_class 10 \
--num_frames 61 \
--num_inference_steps 32 \
--guidance_scale 6.0
```

4 Approach

Task. We construct a compact Human Activity Recognition (HAR) video corpus with seven classes: *Clapping*, *Meet and Split*, *Sitting*, *Standing Still*, *Walking*, *Walking While Reading Book*, and *Walking While Using Phone*. For each class we generate **10** clips (total **70** videos), holding resolution and length fixed (832×480 at 24 fps; 61 frames per clip).

Two prompting regimes. We evaluate two captioning strategies when conditioning the generator:

1. **Simple prompts** (class-focused): minimal, grammatical sentences that name the subject and action only, e.g., “A man is clapping.” or “Two people meet briefly and then split apart.” This encourages the model to produce uncluttered motion aligned to the class label.

2. **Varied prompts** (diversity-focused): short sentences that vary the subject (e.g., man/woman/child/student), occasionally rephrase the action, and lightly vary context words. These small lexical changes promote intra-class diversity while keeping the action semantics intact.

Prompt sets. The exact prompts used for every clip are recorded at generation time and stored in two metadata files: `metadata_simple.csv` (simple regime) and `metadata.csv` (varied regime). For transparency and reproducibility, we include the full lists as tables in the appendix: see *Prompt List (Simple)* and *Prompt List (Varied)*, which enumerate (class, index, prompt) for all 70 videos.¹³

Per-class sampling. For each class $c \in \mathcal{C}$, we generate clips with indices $i = 0 \dots 9$. Each sample is deterministically identified by its row in the metadata (class, index, caption, seed, and settings), allowing exact regeneration. The two regimes are kept separate to isolate the effect of caption complexity on motion fidelity and diversity.

Outputs and bookkeeping. Every video is saved under its class directory, with a sidecar `.txt` containing the exact prompt and the key parameters (*frames*, *steps*, *guidance*, *size*); a master CSV aggregates all clips for quick analysis. For qualitative figures in the paper, we extract one representative frame per clip.

Rationale. Simple captions enforce clean action priors and reduce prompt-induced artifacts; varied captions inject subject diversity and mild lexical variation to prevent near-duplicate clips within each class while preserving the intended action label.

Prompt List (Simple)

Class	Idx	Prompt
Clapping	0	A teenager is clapping.
Clapping	1	A girl is clapping.
Clapping	2	A student is clapping.
Clapping	3	A student is clapping.
Clapping	4	A girl is clapping.
Clapping	5	A teenager is clapping.
Clapping	6	A student is clapping.
Clapping	7	A student is clapping.
Clapping	8	A woman is clapping.
Clapping	9	A girl is clapping.
Meet and Split	0	Two students meet briefly and then split apart.
Meet and Split	1	Two friends meet briefly and then split apart.
Meet and Split	2	Two friends meet briefly and then split apart.
Meet and Split	3	Two strangers meet briefly and then split apart.
Meet and Split	4	Two friends meet briefly and then split apart.
Meet and Split	5	Two students meet briefly and then split apart.
Meet and Split	6	Two students meet briefly and then split apart.
Meet and Split	7	Two strangers meet briefly and then split apart.
Meet and Split	8	Two friends meet briefly and then split apart.
Meet and Split	9	Two friends meet briefly and then split apart.
Sitting	0	A woman is sitting.
Sitting	1	A boy is sitting.
Sitting	2	A boy is sitting.
Sitting	3	A woman is sitting.
Sitting	4	A man is sitting.
Sitting	5	A man is sitting.
Sitting	6	A woman is sitting.
Sitting	7	A teenager is sitting.

¹³The LaTeX for these tables is provided in `prompts_tables.tex`.

Class	Idx	Prompt
Sitting	8	An elderly person is sitting.
Sitting	9	A woman is sitting.
Standing Still	0	A woman is standing still.
Standing Still	1	A boy is standing still.
Standing Still	2	An office worker is standing still.
Standing Still	3	A teenager is standing still.
Standing Still	4	An office worker is standing still.
Standing Still	5	A person is standing still.
Standing Still	6	An office worker is standing still.
Standing Still	7	A student is standing still.
Standing Still	8	A person is standing still.
Standing Still	9	A man is standing still.
Walking	0	A man is walking.
Walking	1	An office worker is walking.
Walking	2	A child is walking.
Walking	3	An elderly person is walking.
Walking	4	An elderly person is walking.
Walking	5	A child is walking.
Walking	6	A teenager is walking.
Walking	7	A girl is walking.
Walking	8	A girl is walking.
Walking	9	A child is walking.
Walking While Reading Book	0	A woman is walking while reading a book.
Walking While Reading Book	1	An elderly person is walking while reading a book.
Walking While Reading Book	2	A student is walking while reading a book.
Walking While Reading Book	3	A person is walking while reading a book.
Walking While Reading Book	4	A woman is walking while reading a book.
Walking While Reading Book	5	A student is walking while reading a book.
Walking While Reading Book	6	An elderly person is walking while reading a book.
Walking While Reading Book	7	A child is walking while reading a book.
Walking While Reading Book	8	A man is walking while reading a book.
Walking While Reading Book	9	A child is walking while reading a book.
Walking While Using Phone	0	A boy is walking while using a phone.
Walking While Using Phone	1	A student is walking while using a phone.
Walking While Using Phone	2	A girl is walking while using a phone.
Walking While Using Phone	3	An elderly person is walking while using a phone.
Walking While Using Phone	4	A child is walking while using a phone.
Walking While Using Phone	5	An office worker is walking while using a phone.
Walking While Using Phone	6	A student is walking while using a phone.

Class	Idx	Prompt
Walking While Using Phone	7	A teenager is walking while using a phone.
Walking While Using Phone	8	A boy is walking while using a phone.
Walking While Using Phone	9	A woman is walking while using a phone.

Prompt List (Varied)

Class	Idx	Prompt
Clapping	0	a tourist claps hands a few times while standing in a train station concourse, at noon. low-angle shot, natural lighting. High quality, realistic motion, 24 fps, (take 1)
Clapping	1	a college student claps hands a few times while standing in a train station concourse, at noon. steady mid-shot, soft shadows. High quality, realistic motion, 24 fps, (take 2)
Clapping	2	a commuter claps hands a few times while standing in a quiet park path, at noon. steady mid-shot, soft shadows. High quality, realistic motion, 24 fps, (take 3)
Clapping	3	a commuter claps hands a few times while standing inside a modern lobby, at noon. static wide shot, realistic textures. High quality, realistic motion, 24 fps, (take 4)
Clapping	4	a middle-aged person claps hands a few times while standing in a train station concourse, in the evening. front-facing tracking shot, realistic textures. High quality, realistic motion, 24 fps, (take 5)
Clapping	5	a commuter claps hands a few times while standing beside a row of trees, in the afternoon. handheld eye-level shot, subtle background motion. High quality, realistic motion, 24 fps, (take 6)
Clapping	6	a jogger claps hands a few times while standing on a pedestrian plaza, in the morning. low-angle shot, slight depth of field. High quality, realistic motion, 24 fps, (take 7)
Clapping	7	an office worker claps hands a few times while standing inside a modern lobby, in the evening. steady mid-shot, realistic textures. High quality, realistic motion, 24 fps, (take 8)
Clapping	8	a young adult claps hands a few times while standing on a pedestrian plaza, at noon. static wide shot, soft shadows. High quality, realistic motion, 24 fps, (take 9)
Clapping	9	a middle-aged person claps hands a few times while standing inside a modern lobby, at noon. static wide shot, slight depth of field. High quality, realistic motion, 24 fps, (take 10)
Meet and Split	0	a commuter meets briefly with another person, then they walk away in opposite directions in a train station concourse, at noon. steady mid-shot, subtle background motion. High quality, realistic motion, 24 fps, (take 1)
Meet and Split	1	a teenager meets briefly with another person, then they walk away in opposite directions near a building entrance, in the afternoon. slight side-angle tracking shot, soft shadows. High quality, realistic motion, 24 fps, (take 2)
Meet and Split	2	a young adult meets briefly with another person, then they walk away in opposite directions on a pedestrian plaza, at noon. static wide shot, realistic textures. High quality, realistic motion, 24 fps, (take 3)

Class	Idx	Prompt
Meet and Split	3	an office worker meets briefly with another person, then they walk away in opposite directions on a campus walkway, in the afternoon. low-angle shot, soft shadows. High quality, realistic motion, 24 fps, (take 4)
Meet and Split	4	an office worker meets briefly with another person, then they walk away in opposite directions on a city sidewalk, at noon. steady mid-shot, subtle background motion. High quality, realistic motion, 24 fps, (take 5)
Meet and Split	5	a visitor meets briefly with another person, then they walk away in opposite directions inside a modern lobby, at golden hour. front-facing tracking shot, soft shadows. High quality, realistic motion, 24 fps, (take 6)
Meet and Split	6	a shopper meets briefly with another person, then they walk away in opposite directions in a bright hallway, at noon. low-angle shot, natural lighting. High quality, realistic motion, 24 fps, (take 7)
Meet and Split	7	a shopper meets briefly with another person, then they walk away in opposite directions on a campus walkway, at noon. front-facing tracking shot, realistic textures. High quality, realistic motion, 24 fps, (take 8)
Meet and Split	8	a middle-aged person meets briefly with another person, then they walk away in opposite directions on a campus walkway, at golden hour. front-facing tracking shot, realistic textures. High quality, realistic motion, 24 fps, (take 9)
Meet and Split	9	a jogger meets briefly with another person, then they walk away in opposite directions in a quiet park path, in the afternoon. static wide shot, soft shadows. High quality, realistic motion, 24 fps, (take 10)
Sitting	0	a jogger sits down calmly on a bench and stays seated inside a modern lobby, in the morning. slight side-angle tracking shot, slight depth of field. High quality, realistic motion, 24 fps, (take 1)
Sitting	1	an office worker sits down calmly on a bench and stays seated on a city sidewalk, at golden hour. static wide shot, natural lighting. High quality, realistic motion, 24 fps, (take 2)
Sitting	2	an office worker sits down calmly on a bench and stays seated on a pedestrian plaza, in the afternoon. steady mid-shot, slight depth of field. High quality, realistic motion, 24 fps, (take 3)
Sitting	3	a middle-aged person sits down calmly on a bench and stays seated in a quiet park path, in the afternoon. slight side-angle tracking shot, subtle background motion. High quality, realistic motion, 24 fps, (take 4)
Sitting	4	a college student sits down calmly on a bench and stays seated near a building entrance, in the evening. slight side-angle tracking shot, natural lighting. High quality, realistic motion, 24 fps, (take 5)
Sitting	5	a teenager sits down calmly on a bench and stays seated on a campus walkway, in the afternoon. low-angle shot, subtle background motion. High quality, realistic motion, 24 fps, (take 6)
Sitting	6	a tourist sits down calmly on a bench and stays seated on a wide corridor with windows, in the afternoon. static wide shot, slight depth of field. High quality, realistic motion, 24 fps, (take 7)
Sitting	7	a college student sits down calmly on a bench and stays seated in a bright hallway, in the afternoon. low-angle shot, natural lighting. High quality, realistic motion, 24 fps, (take 8)
Sitting	8	a young adult sits down calmly on a bench and stays seated in a train station concourse, in the evening. low-angle shot, subtle background motion. High quality, realistic motion, 24 fps, (take 9)
Sitting	9	a jogger sits down calmly on a bench and stays seated on a city sidewalk, in the afternoon. low-angle shot, realistic textures. High quality, realistic motion, 24 fps, (take 10)

Class	Idx	Prompt
Standing Still	0	a jogger stands still for a moment, looking ahead on a wide corridor with windows, in the afternoon. low-angle shot, slight depth of field. High quality, realistic motion, 24 fps, (take 1)
Standing Still	1	a college student stands still for a moment, looking ahead on a pedestrian plaza, in the afternoon. front-facing tracking shot, natural lighting. High quality, realistic motion, 24 fps, (take 2)
Standing Still	2	a jogger stands still for a moment, looking ahead near a building entrance, in the afternoon. steady mid-shot, subtle background motion. High quality, realistic motion, 24 fps, (take 3)
Standing Still	3	a middle-aged person stands still for a moment, looking ahead on a wide corridor with windows, at golden hour. front-facing tracking shot, natural lighting. High quality, realistic motion, 24 fps, (take 4)
Standing Still	4	a teenager stands still for a moment, looking ahead on a wide corridor with windows, in the morning. slight side-angle tracking shot, natural lighting. High quality, realistic motion, 24 fps, (take 5)
Standing Still	5	a commuter stands still for a moment, looking ahead on a wide corridor with windows, in the afternoon. steady mid-shot, subtle background motion. High quality, realistic motion, 24 fps, (take 6)
Standing Still	6	an office worker stands still for a moment, looking ahead near a building entrance, in the morning. slight side-angle tracking shot, natural lighting. High quality, realistic motion, 24 fps, (take 7)
Standing Still	7	a jogger stands still for a moment, looking ahead in a train station concourse, in the afternoon. handheld eye-level shot, subtle background motion. High quality, realistic motion, 24 fps, (take 8)
Standing Still	8	a tourist stands still for a moment, looking ahead in a bright hallway, at golden hour. low-angle shot, realistic textures. High quality, realistic motion, 24 fps, (take 9)
Standing Still	9	a jogger stands still for a moment, looking ahead in a train station concourse, in the morning. low-angle shot, slight depth of field. High quality, realistic motion, 24 fps, (take 10)
Walking	0	a visitor walks forward at a normal pace on a wide corridor with windows, in the morning. slight side-angle tracking shot, natural lighting. High quality, realistic motion, 24 fps, (take 1)
Walking	1	a young adult walks forward at a normal pace in a train station concourse, at golden hour. static wide shot, natural lighting. High quality, realistic motion, 24 fps, (take 2)
Walking	2	a tourist walks forward at a normal pace in a bright hallway, at noon. slight side-angle tracking shot, soft shadows. High quality, realistic motion, 24 fps, (take 3)
Walking	3	a commuter walks forward at a normal pace near a building entrance, in the afternoon. static wide shot, realistic textures. High quality, realistic motion, 24 fps, (take 4)
Walking	4	a jogger walks forward at a normal pace inside a modern lobby, at noon. steady mid-shot, soft shadows. High quality, realistic motion, 24 fps, (take 5)
Walking	5	a teenager walks forward at a normal pace on a city sidewalk, at noon. static wide shot, realistic textures. High quality, realistic motion, 24 fps, (take 6)
Walking	6	a visitor walks forward at a normal pace in a bright hallway, in the evening. static wide shot, subtle background motion. High quality, realistic motion, 24 fps, (take 7)
Walking	7	a teenager walks forward at a normal pace in a bright hallway, in the evening. static wide shot, soft shadows. High quality, realistic motion, 24 fps, (take 8)
Walking	8	a shopper walks forward at a normal pace on a city sidewalk, at golden hour. front-facing tracking shot, subtle background motion. High quality, realistic motion, 24 fps, (take 9)

Class			Idx	Prompt
Walking			9	a tourist walks forward at a normal pace on a pedestrian plaza, in the evening. slight side-angle tracking shot, realistic textures. High quality, realistic motion, 24 fps, (take 10)
Walking While Reading Book			0	a college student walks slowly while reading a small book on a wide corridor with windows, at noon. steady mid-shot, realistic textures. High quality, realistic motion, 24 fps, (take 1)
Walking While Reading Book			1	a commuter walks slowly while reading a small book in a quiet park path, in the evening. slight side-angle tracking shot, slight depth of field. High quality, realistic motion, 24 fps, (take 2)
Walking While Reading Book			2	a shopper walks slowly while reading a small book near a building entrance, in the evening. low-angle shot, natural lighting. High quality, realistic motion, 24 fps, (take 3)
Walking While Reading Book			3	a young adult walks slowly while reading a small book in a bright hallway, in the afternoon. static wide shot, slight depth of field. High quality, realistic motion, 24 fps, (take 4)
Walking While Reading Book			4	a middle-aged person walks slowly while reading a small book on a city sidewalk, in the evening. static wide shot, realistic textures. High quality, realistic motion, 24 fps, (take 5)
Walking While Reading Book			5	an office worker walks slowly while reading a small book beside a row of trees, in the morning. handheld eye-level shot, realistic textures. High quality, realistic motion, 24 fps, (take 6)
Walking While Reading Book			6	a middle-aged person walks slowly while reading a small book on a city sidewalk, at golden hour. steady mid-shot, realistic textures. High quality, realistic motion, 24 fps, (take 7)
Walking While Reading Book			7	a visitor walks slowly while reading a small book in a train station concourse, in the evening. handheld eye-level shot, subtle background motion. High quality, realistic motion, 24 fps, (take 8)
Walking While Reading Book			8	a middle-aged person walks slowly while reading a small book inside a modern lobby, in the afternoon. front-facing tracking shot, slight depth of field. High quality, realistic motion, 24 fps, (take 9)
Walking While Reading Book			9	a middle-aged person walks slowly while reading a small book near a building entrance, in the evening. front-facing tracking shot, realistic textures. High quality, realistic motion, 24 fps, (take 10)
Walking While Using Phone			0	a middle-aged person walks while looking at a phone in one hand in a bright hallway, at noon. static wide shot, subtle background motion. High quality, realistic motion, 24 fps, (take 1)
Walking While Using Phone			1	a teenager walks while looking at a phone in one hand on a city sidewalk, in the evening. static wide shot, realistic textures. High quality, realistic motion, 24 fps, (take 2)
Walking While Using Phone			2	a shopper walks while looking at a phone in one hand on a campus walkway, in the morning. steady mid-shot, subtle background motion. High quality, realistic motion, 24 fps, (take 3)
Walking While Using Phone			3	a teenager walks while looking at a phone in one hand on a campus walkway, in the afternoon. low-angle shot, realistic textures. High quality, realistic motion, 24 fps, (take 4)
Walking While Using Phone			4	a teenager walks while looking at a phone in one hand on a pedestrian plaza, in the morning. handheld eye-level shot, realistic textures. High quality, realistic motion, 24 fps, (take 5)
Walking While Using Phone			5	a teenager walks while looking at a phone in one hand on a pedestrian plaza, at noon. slight side-angle tracking shot, soft shadows. High quality, realistic motion, 24 fps, (take 6)
Walking While Using Phone			6	an office worker walks while looking at a phone in one hand on a pedestrian plaza, at noon. static wide shot, soft shadows. High quality, realistic motion, 24 fps, (take 7)
Walking While Using Phone			7	a tourist walks while looking at a phone in one hand in a bright hallway, at golden hour. handheld eye-level shot, soft shadows. High quality, realistic motion, 24 fps, (take 8)

Class			Idx	Prompt
Walking While Using Phone			8	an office worker walks while looking at a phone in one hand near a building entrance, at golden hour. steady mid-shot, realistic textures. High quality, realistic motion, 24 fps, (take 9)
Walking While Using Phone			9	a shopper walks while looking at a phone in one hand beside a row of trees, at golden hour. handheld eye-level shot, subtle background motion. High quality, realistic motion, 24 fps, (take 10)